

UPPSALA UNIVERSITY



MEDICAL INFORMATICS

1MD026 / 1MD030

Analysis of the Patient Discharge method in COSMIC EHR

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September 15, 2026

Abstract

This project centered on analyzing the discharge process of the COSMIC Electronic Health Record (EHR) to identify pain points in the current clinical workflow and propose improvements to enhance automation, usability, and overall efficiency. Drawing on a literature review to establish foundational best practices and a clinical survey/analysis to capture real-world challenges, the study revealed that while COSMIC is a valuable tool, it does not fully support an efficient and safe workflow. Key findings highlighted that the discharge process is time-consuming and prone to missed steps due to excessively long care plans, suffers from duplicate documentation and efficiency loss, and is hampered by communication gaps stemming from the parallel use of COSMIC Messenger and COSMIC Link. Furthermore, unclear professional responsibilities often cause staff frustration and delays during multidisciplinary involvement. Improvements, guided by Nielsen’s usability heuristics, include redesigning the interface into a concise, checklist-based layout, standardizing communication, integrating functions like “Status for general nursing” directly into COSMIC Link, and developing a shared “Discharge Dashboard” to improve coordination. The most ambitious solution is the implementation of a structured discharge summary template complemented by security measures like Role-Based Access Control (RBAC) to ensure data consistency and confidentiality.

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1 Background

1.1 Introduction

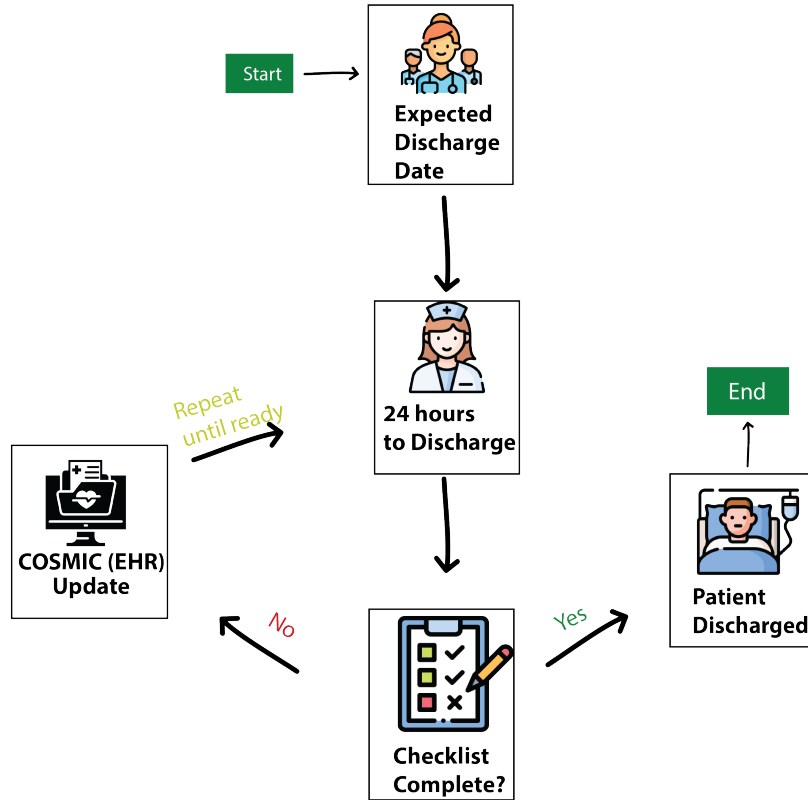


Figure 1: Illustration of patient discharge (icons source: Flaticon)

A discharge (as shown in Figure 1) is a formal process of releasing a patient from hospital care. A discharge summary is a document that captures a patient's hospital stay, treatment changes, and plans for post-discharge care. The discharge of a patient from the hospital to home or outpatient care is critical to patient safety and quality of care. More often than not, patients receive complex instructions about medications, follow-up appointments, and lifestyle changes, which lead to a lack of understanding and remembrance, poor adherence, complications, and sometimes preventable hospital readmissions. These problems can be improved with better communication [1].

Electronic Health Records (EHRs) are digital versions of a patient’s medical history used to support clinical care and coordination. EHRs are capable of managing discharges by generating discharge summaries. One of the most widely deployed EHR systems in Sweden is COSMIC (developed by Cambio and shown in Figure 2) [2], used in nearly all departments at Akademiska sjukhuset. EHR-based discharge processes can face usability challenges, despite their potential benefits, as shown in a Swedish usability study [3] of COSMIC, where clinicians often struggled with the medication module, slow system performance, and inefficiencies in locating relevant patient data. Our project centers on analyzing the **discharge process of COSMIC**. We analyze how clinicians prepare discharge summaries, identify pain points in the current workflow, and propose improvements to enhance automation, usability, and overall efficiency.

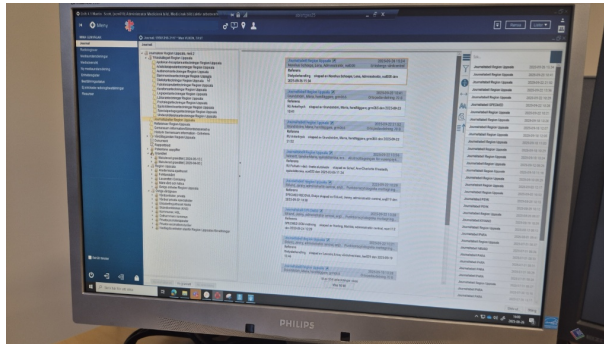


Figure 2: A photo of the COSMIC EHR screen taken during our visit to Akademiska Sjukhuset

1.2 Literature Review

Apart from improved communication [1], this literature review highlights existing challenges that have been addressed to ensure a smoother patient experience, with efficiency and safety in mind.

Firstly, discharge summaries vary greatly in format and content, which increases inconsistencies. Weak or inconsistent discharge documents are a consistent risk factor in transitions of care [4]. To address this, one of the recurrent themes in literature is that the standardization of discharge summaries can significantly improve consistency and overall discharge quality. For instance, the authors [5] describe how a hospital implemented a template within the EHR, achieving approximately 90% adoption across clinicians in 18 months, with recipients reporting a higher quality of information. Similarly, the authors of [6] aimed to enhance the consistency and quality of discharge summaries on busy hospital wards, resulting in a 40% increase in compliance scores from 55% to 95%. The issue of consistency is more significant when patients transfer between hospitals that use different electronic health record (EHR) systems. Each system stores information using its

own unique rules and methods, requiring interoperability. The authors of [7] developed an automated method to translate clinical semantics between the proprietary COSMIC templates and standardized openEHR archetypes, demonstrating a feasible solution to this problem.

Secondly, medication errors are a frequent problem during discharge, which can cause serious patient harm, lead to readmissions, and increase healthcare costs. Pharmacists identified medication errors in 68% of reviewed patients, with 75% of these errors having serious potential harm [8]. These findings strongly support integrating pharmacist reviews into the discharge workflow as part of the COSMIC redesign.

Furthermore, automation and patient-centered processes are key to quality discharge. Patients rarely understand technical medical language; yet, they must follow discharge plans. The authors [9] found that edits to discharge summaries (shorter sentences, clearer language, and avoidance of unexplained abbreviations) led to improved usability and comprehension among clinicians and patients. The authors of [10] used a Large Language Model (LLM) to convert technical summaries into more readable text, where the AI was able to create patient-friendly discharge summaries that were significantly more readable and understandable (81% score vs. 13% for the original). Though the use of AI is helpful for accessibility, it must include physician reviews before deployment to ensure accuracy and patient safety. Finally, the authors of [11] showed that a remote "virtual discharge nurse," who used EHR tools, successfully increased the percentage of discharges before 2 PM; however, it did not reduce the total length of stay.

2 Problem Description

While the literature provides insights into best practices, the survey (in Appendix B) reveals the actual **challenges faced by clinicians in practice during the discharge process**, primarily at Akademiska Sjukhuset. We received 4 responses in total: 2 from nurses, 1 from an assistant nurse, and 1 from an unknown respondent within the department who is likely related to operating the Cosmic EHR.

Respondents noted that the standard care plan, particularly for coordination cases, is excessively long with too many search terms, making the discharge process time-consuming, difficult to manage, and prone to missed steps. A lot of workflow problems and inconsistencies highlighted in the background section were actually faced in reality. Staff responsible for discharging patients must first open the "Status" tab in the patient record to retrieve the relevant search terms and then manually transfer this information into COSMIC Link, where the discharge and the individual care plan are managed. Respondents reported that sharing information (receiving caregiver, discharge time) often requires duplicate entry in both COSMIC Link's general message section and the discharge care plan, causing extra work and inefficiency. The survey also revealed that parallel use of COSMIC Messenger and COSMIC Link for communication creates confu-

sion and inefficiency, as staff must constantly switch between tools to keep information accurate and up to date. Multidisciplinary involvement in patient discharge, where several professional groups must provide assessments, often makes the process lengthy and inefficient, especially when documentation and communication are not continuous. The survey indicated that unclear professional responsibilities in the discharge process can cause staff frustration and delays when workflow steps are stalled or incomplete. The survey also revealed that nurses and assistant nurses perceive many physicians struggle with discharge summary notes due to limited knowledge or uncertainty, causing delays that prolong the discharge process and increase staff frustration.

In summary, the survey indicates that while COSMIC is a valuable tool for patient discharge, it does not fully support an efficient and safe workflow, with key challenges including time consumption, duplicate documentation, communication gaps, and unclear role distribution. Improving care planning and information transfer could enhance both efficiency and patient safety.

3 Proposed Improvements

Based on the shortcomings identified in the discharge summary workflow within the COSMIC subsystem, several improvements are proposed to enhance the usability and efficiency of the system, to support its users, and ensure better continuity of care for patients. These improvements are guided by Nielsen’s usability heuristics, providing a structured framework to enhance workflow efficiency for users of the COSMIC EHR. Firstly, the interface should be redesigned into a concise, checklist-based layout (shown in Figure 3) to reduce cognitive load, improve navigation, and guide users through each discharge step efficiently. [12] found that less display fragmentation and reduced data entry requirements reduced users’ cognitive load, decreased confusion, and improved system usability and satisfaction. These improvements follow Nielsen’s heuristics of “Aesthetic and minimalist design” and “Recognition rather than recall” [13]. A more detailed illustration is shown in Figure 4 in Appendix A.

Discharge Summary Checklist

- ☐ Medication list
- ☐ Follow-up appointments
- ☐ Transportation arranged
- ☐ Dietary restrictions noted
- ☐ Pain management plan
- ☐ Wound care instructions
- ☐ Prescriptions filled
- ☐ Emergency contact information
- ☐ Instructions for care

Figure 3: Proposed Discharge Summary Checklist

An excellent idea is to integrate the “Status for general nursing” function directly into COSMIC Link, allowing users to update patient status without leaving the discharge interface. The “Status for General Nursing” acts as a summary overview that is mirrored into the Case View in Cosmic Link under the Journal tab, reflecting the latest entries documented under specific search terms (such as nutrition, mobility, and communication) within the patient’s core Cosmic journal. The integration ensures automatic synchronization with the medical record, reducing manual work and keeping information accurate and consistent across the system. Additionally, all discharge communication should be standardized within COSMIC Link to reduce confusion and duplicate documentation. This creates a single, consistent environment for multidisciplinary coordination, lowering cognitive load and improving task efficiency [12]. Both of these improvements are illustrated in Figure 5 in Appendix A. The suggested improvement aligns with Nielsen’s 4th usability heuristic, “Consistency and Standards,” which emphasizes uniform interfaces, as well as the 5th heuristic, “Error Prevention,” by reducing redundant data entry and inconsistencies [13]. Implementing a structured discharge summary template within COSMIC with clearly defined sections can improve both documentation quality and task completion time (Figure 7 in Appendix A). This standardized template promotes consistency, reduces ambiguity, ensures all essential details are documented, and uses visible section labels to guide users through each step rather than relying on memory [14]. It can also be complemented by measures like **role-based access control (RBAC)** and **smart data-sharing mechanisms** to manage data sharing despite strict **confidential-**

ity requirements [15]. The suggested solution aligns with Nielsen’s 4th and 6th usability heuristics, “Consistency and Standards” and “Recognition Rather Than Recall” [13].

Regarding professional role clarity, it is recommended to adopt role-specific training programs and provide an internal guide within the COSMIC system to clarify professional responsibilities. In practice, hospitals should actively collect feedback from different professionals on how COSMIC Link functions in daily use. This helps identify confusing or time-consuming aspects, enabling targeted improvements and allowing vendors to update the system accordingly [16]. The suggested improvements are supported by Nielsen’s 10th usability heuristic, “Help and Documentation” about accessible and context-sensitive help and guidance [13].

Finally, to overcome the challenges posed by multidisciplinary involvement, it is recommended to develop a shared “Discharge Dashboard” within COSMIC Link to improve coordination and workflow efficiency (Figure 6 in Appendix A). In practice, the dashboard would display all discharge tasks, their completion status, and the responsible professionals, using color-coded indicators for easy tracking. This setup allows parallel workflow, lets users update their sections directly, and reduces waiting times between roles [17]. The suggested improvement aligns with Nielsen’s 1st heuristic, “Visibility of system status,” about keeping users informed about what is going on through appropriate feedback [13].

4 Discussion

With the groundwork established through literature review, surveys, and system evaluation, it is now useful to reflect on the project process itself - what worked effectively, what could have been approached differently, and what knowledge emerged from the experience. Our approaches offered a comprehensive understanding of how discharge summaries are prepared and the key shortcomings and improvements for efficient and safe workflows. The literature review provided a foundational understanding of discharge workflows, highlighting best practices and common challenges observed in different EHRs. The survey captured users’ real-world experiences, revealing practical pain points such as long care plans, duplicate documentation, and communication inefficiencies. Additionally, the IT system evaluation helped identify structural and workflow-related limitations within COSMIC that contribute to these challenges.

Regarding the overall project work, the surveying process was the aspect that took the longest time, due to delays in communication from the hospital staff. However, since the hospital staff are often occupied with clinical duties and patient care, this delay was expected. Several alternative methods were suggested during our group discussions, such as interviewing staff in person and using online interview data on COSMIC from previous research papers, but we used surveying since it proved to be the most authentic methodology given our time frame, generating a variety of answers which were then used to enhance the quality of our suggestions. While the recent surge in AI further raises

questions about the applicability and reliability of AI-powered solutions in discharge processes, current evidence indicates that artificial intelligence (AI) can assist in generating hospital discharge summaries from inpatient records, but complete automation is not yet feasible [18] unless a thorough human-in-the-loop verification is in place [19]. More research and solution proposals on this aspect were beyond our time scope and could be explored with additional time or in a future project phase.

5 Conclusion

The most important takeaway from the analysis is that in an Electronic Health Record (EHR) system like COSMIC, it is crucial to ensure that the discharge workflow supports both efficiency and patient safety. The findings from our survey analysis, IT system evaluation, and literature review emphasize the need for usability, consistency, and patient-centred design to reduce redundant steps and enhance multidisciplinary collaboration. Clear role distribution, unified communication tools, and an intuitive interface are essential to prevent errors and delays in discharge preparation. The analysis highlights that EHRs should not only serve as information repositories but also actively facilitate seamless, coordinated, and user-friendly discharge processes aligned with clinical practice.

The most ambitious solution in our analysis is implementing a **structured discharge summary template**, as it requires complex integrations and security measures like **RBAC** and **smart data-sharing mechanisms** to ensure confidentiality. This standardization is crucial for addressing ambiguity but demands a significant technical overhaul. By starting with simpler improvements such as integrating the “Status for general nursing” function into COSMIC Link, standardizing discharge communication, and providing internal guides or role-specific training, we can begin streamlining workflows and move toward the larger goal effectively. Overall, these improvements will help achieve the broad goals of efficient, safe, and ethically responsible patient care by supporting clear workflows, accurate documentation, and secure information sharing.

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A Example User Interfaces: Improvements

The images below are example illustrations that show how user interfaces could potentially be designed to solve the problems discussed in the report. Note that the data shown in the illustrations is synthetically generated and is not representative of any aspects of the real world. The illustrations are generated by Claude Artifacts[20] and serve as a starting point for further implementation.

Discharge Coordination Checklist

Progress: 5/8 completed

- ☒ **Patient Information Verified**
 - Demographics
 - Contact details
 - Insurance
- ☒ **Medical Documentation Complete**
 - Diagnoses
 - Medications
 - Allergies
- ☒ **Discharge Medications Reviewed**
 - Prescriptions
 - Instructions
 - Pharmacy contacted
- ☒ **Follow-up Appointments Scheduled**
 - Primary care
 - Specialists
 - Lab work
- ☒ **Home Care Plan Established**
 - Services arranged
 - Equipment ordered
 - Caregiver trained
- ☐ **Patient Education Completed**
 - Condition explained
 - Warning signs reviewed
 - Materials provided
- ☐ **Transportation Arranged**
 - Discharge ride
 - Future appointments
 - Emergency access
- ☐ **Final Coordination Sign-off**
 - All tasks verified
 - Patient/family confirmed
 - Documentation filed

Figure 4: Conceptual Illustration of the Proposed Checklist-Based Discharge Interface

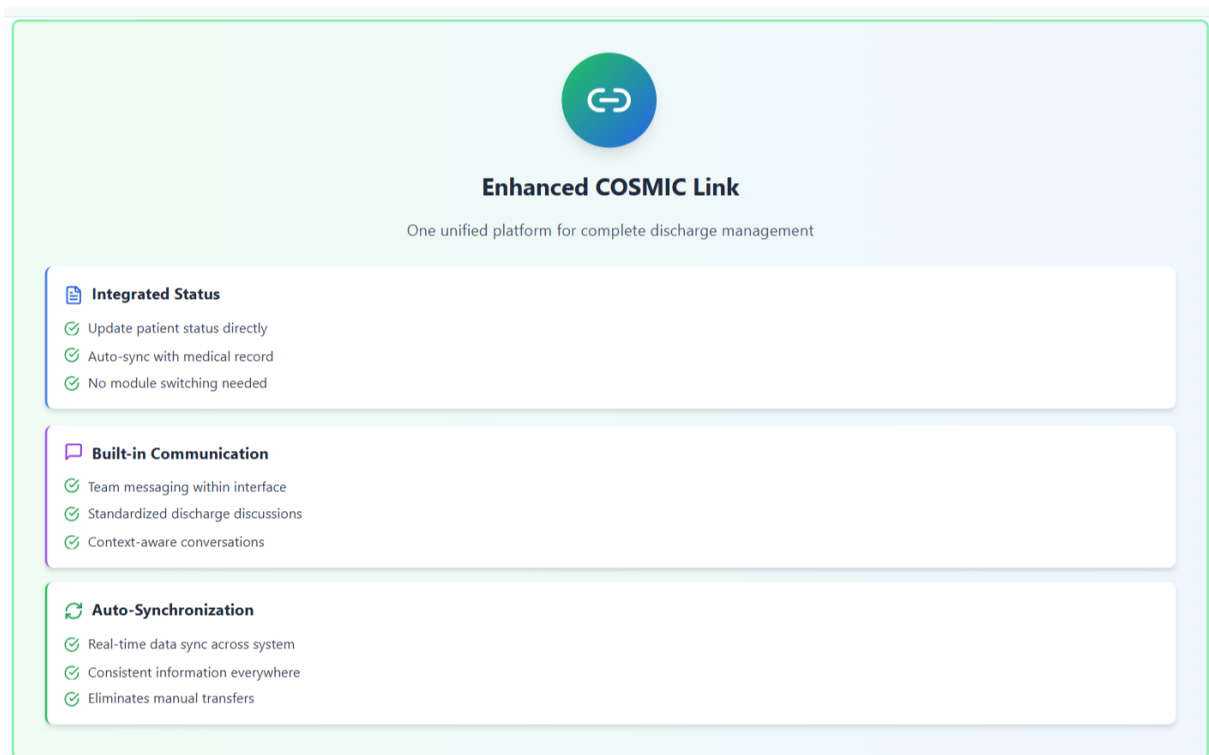


Figure 5: Conceptual Illustration showing the Extend Cosmic Link with the proposed feature additions

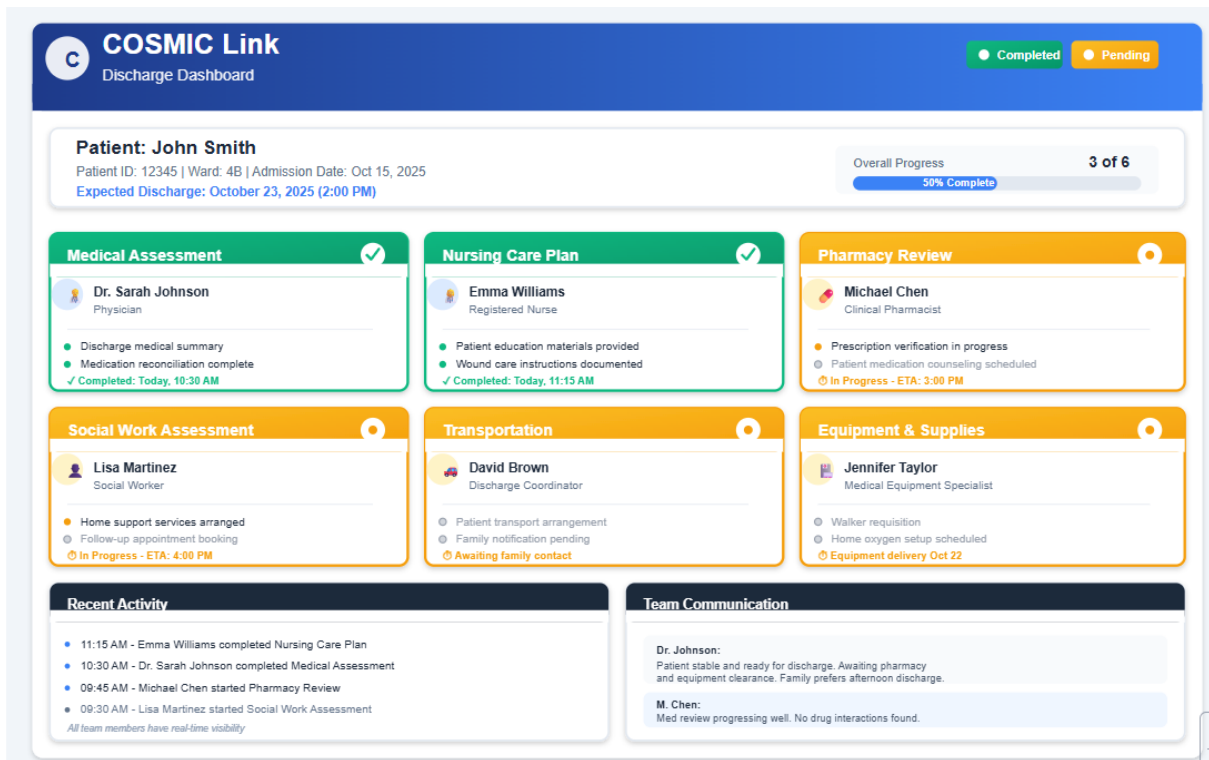


Figure 6: Illustration of the proposed Shared Discharge Dashboard

Discharge Summary Template
25% Complete

Patient: John Doe (MRN: 123456) • Room: 302B

Progress: 2 of 8 sections

1. Patient Demographics
Basic patient identification information
Completed

Name: John Doe
DOB: 05/15/1957 (68 years)
MRN: 123456
Gender: Male

2. Admission & Discharge Dates
Timeline of hospital stay
Completed

Admission Date: October 10, 2025
Discharge Date: October 18, 2025
Length of Stay: 8 days
Admission Type: Emergency

3. Primary & Secondary Diagnoses
Main conditions treated during hospitalization
Required

Click to complete this section...

4. Procedures & Interventions
Treatments performed during hospitalization

Optional - Add if procedures were performed...

Figure 7: Conceptual Illustration of the proposed Structured Discharge Template

B Survey Responses

Below are the collected responses from the user survey. The original report is 10 pages, excluding the appendix. The responses are both in Swedish and English.

COSMIC – användarundersökning

Syftet med denna enkät är att samla in erfarenheter och synpunkter kring utskrivningsprocessen i journalsystemet COSMIC. Svaren är anonyma och kommer enbart att användas för studieändamål. Enkäten tar bara några minuter att besvara. Tack för din medverkan!

Vilken yrkesroll har du?

- ☐ Läkare
- ☐ Sjuksköterska
- ☐ IT - support
- ☐ Farmaceut
- ☐ Other: _____

Hur länge har du arbetat med COSMIC?

- ☐ < 1 år
- ☒ 1 - 3 år
- ☐ 4 - 6 år
- ☐ > 6 år

Hur ofta använder du COSMIC?

- ☒ Vardagligen
- ☐ Några gånger i veckan
- ☐ Sällan

Hur upplever du att utskrivningsprocessen fungerar i COSMIC idag?

- ☐ Mycket bra
- ☒ Ganska bra
- ☐ Varken eller
- ☐ Ganska dåligt
- ☐ Mycket dåligt

I vilken utsträckning använder du COSMIC under utskrivningsprocessen av en patient?

- ☐ Hela processen sker i COSMIC
- ☒ Vissa delar görs i andra system
- ☐ Mest på papper eller via telefon

Vilket/vilka moment gör du i COSMIC vid utskrivning av patient?

- ☐ Skriver epikris
- ☐ Uppdaterar vårdplan
- ☒ Skickar vårdbegäran / SVPL
- ☐ Hanterar läkemedelslista
- ☐ Other:

Hur lång tid tar utskrivningen vanligtvis att genomföra i COSMIC?

-

Är det lätt att hitta och använda rätt funktioner i COSMIC vid utskrivning (t.ex. epikris, vårdplan, läkemedel)?

- ☒ Ja, alltid.
- ☐ Oftast
- ☐ Sällan
- ☐ Nej det är svårt

Hur ofta behöver du be om hjälp från kollegor eller IT-support vid utskrivningar i COSMIC?

- ☐ Aldrig
- ☒ Ibland
- ☐ Ofta
- ☐ Nästan alltid

Har du upplevt att patienter fått fel eller ofullständig information vid utskrivning?

- ☒ Ja
- ☐ Nej
- ☐ Vet inte

Om ja på tidigare fråga, vad berodde det på?

Oftast missförstånd eller misskommunikation

Har du upplevt dubbelarbete (t.ex. behövt skriva in samma sak i flera system)?

- ☒ Ja
- ☐ Nej
- ☐ Vet inte

Vilka moment upplever du som mest tidskrävande vid utskrivningsprocessen i COSMIC?

Upplever du något problem med COSMIC vid utskrivningsprocessen, i så fall vad?

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COSMIC – användarundersökning

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Vilken yrkesroll har du?

- ☐ Läkare
- ☒ Sjuksköterska
- ☐ IT - support
- ☐ Farmaceut
- ☐ Other: _____

Hur länge har du arbetat med COSMIC?

- ☐ < 1 år
- ☐ 1 - 3 år
- ☐ 4 - 6 år
- ☒ > 6 år

Hur ofta använder du COSMIC?

- ☒ Vardagligen
- ☐ Några gånger i veckan
- ☐ Sällan

Hur upplever du att utskrivningsprocessen fungerar i COSMIC idag?

- ☐ Mycket bra
- ☒ Ganska bra
- ☐ Varken eller
- ☐ Ganska dåligt
- ☐ Mycket dåligt

I vilken utsträckning använder du COSMIC under utskrivningsprocessen av en patient?

- ☐ Hela processen sker i COSMIC
- ☒ Vissa delar görs i andra system
- ☐ Mest på papper eller via telefon

Vilket/vilka moment gör du i COSMIC vid utskrivning av patient?

- ☒ Skriver epikris
- ☒ Uppdaterar vårdplan
- ☐ Skickar vårbegäran / SVPL
- ☒ Hanterar läkemedelslista
- ☒ Other: Kommunicerar med kommunen/privata aktörer/vårdcentralen

Hur lång tid tar utskrivningen vanligtvis att genomföra i COSMIC?

Svårt att specificera. Utskrivningsprocessen pågår från att patienten blir inskriven i Cosmic Link tills att patienten går hem. Vi har kontinuerligt kontakt med kommunen och övriga aktörer, uppdaterar status så att övriga aktörer kan följa patienten etc. På utskrivningsdagen beror tiden på hur mycket som är förberett sedan tidigare. Själva utskrivningen från beläggningslistan är inte tidskrävande.

Är det lätt att hitta och använda rätt funktioner i COSMIC vid utskrivning (t.ex. epikris, vårdplan, läkemedel)?

- ☐ Ja, alltid.
- ☒ Oftast
- ☐ Sällan
- ☐ Nej det är svårt

Hur ofta behöver du be om hjälp från kollegor eller IT-support vid utskrivningar i COSMIC?

- ☒ Aldrig
- ☐ Ibland
- ☐ Ofta
- ☐ Nästan alltid

Har du upplevt att patienter fått fel eller ofullständig information vid utskrivning?

- ☒ Ja
- ☐ Nej
- ☐ Vet inte

Om ja på tidigare fråga, vad berodde det på?

Ofta bristande info från läkare.

Har du upplevt dubbelarbete (t.ex. behövt skriva in samma sak i flera system)?

- ☒ Ja
- ☐ Nej
- ☐ Vet inte

Vilka moment upplever du som mest tidskrävande vid utskrivningsprocessen i COSMIC?

Svårt att svara på era frågor då det är lite oklart om ni syftar på cosmic i sig eller cosmic Link som främst används för utskrivningsprocessen. Det är i Cosmic Link som det mesta av utskrivningsprocessen sker. Om man kontinuerligt följer utskrivningsprocessen framåt under vårdtiden och alla gör sin del är inte utskrivningsprocessen så tidskrävande. Det handlar mer om riktlinjer, utformning av anteckningar/rubriker och sekretess som gör att utskrivningsprocessen blir omständigt. Många läkare har även svårt för utskrivningsklaranteckningen vilket gör att det kan släpa efter. Det handlar mer om innehållet i anteckningen samt okunskap.

Upplever du något problem med COSMIC vid utskrivningsprocessen, i så fall vad?

Det hade varit lättare om man till exempel kunde uppdatera Status för allmän omvårdnad direkt i Cosmic Link istället för att gå in i Status i journalen och hämta upp alla sökord. Det känns också som dubbelarbete att dokumentera/informera i både generella meddelanden i Cosmic Link samt i den generella vårdplanen för utskrivning. Till exempel bokad transport, vem möter patienten, vilken tid, vilka hjälpmedel skickas med osv. Vissa aktörer använder fortfarande Cosmic messenger som en del i utskrivningsprocessen vilket bara blir rörigt och dubbelarbete då det är enklare om alla kommunicerar enbart i Cosmic Link.

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COSMIC – användarundersökning

Syftet med denna enkät är att samla in erfarenheter och synpunkter kring utskrivningsprocessen i journalsystemet COSMIC. Svaren är anonyma och kommer enbart att användas för studieändamål. Enkäten tar bara några minuter att besvara. Tack för din medverkan!

Vilken yrkesroll har du?

- ☐ Läkare
- ☐ Sjuksköterska
- ☐ IT - support
- ☐ Farmaceut
- ☒ Other: Undersköterska

Hur länge har du arbetat med COSMIC?

- ☐ < 1 år
- ☐ 1 - 3 år
- ☐ 4 - 6 år
- ☒ > 6 år

Hur ofta använder du COSMIC?

- ☒ Vardagligen
- ☐ Några gånger i veckan
- ☐ Sällan

Hur upplever du att utskrivningsprocessen fungerar i COSMIC idag?

- ☐ Mycket bra
- ☒ Ganska bra
- ☐ Varken eller
- ☐ Ganska dåligt
- ☐ Mycket dåligt

I vilken utsträckning använder du COSMIC under utskrivningsprocessen av en patient?

- ☒ Hela processen sker i COSMIC
- ☐ Vissa delar görs i andra system
- ☐ Mest på papper eller via telefon

Vilket/vilka moment gör du i COSMIC vid utskrivning av patient?

- ☐ Skriver epikris
- ☒ Uppdaterar vårdplan
- ☒ Skickar vårdbegäran / SVPL
- ☐ Hanterar läkemedelslista
- ☒ Other: det mesta,

Hur lång tid tar utskrivningen vanligtvis att genomföra i COSMIC?

.....

Är det lätt att hitta och använda rätt funktioner i COSMIC vid utskrivning (t.ex. epikris, vårdplan, läkemedel)?

- ☐ Ja, alltid.
- ☒ Oftast
- ☐ Sällan
- ☐ Nej det är svårt

Hur ofta behöver du be om hjälp från kollegor eller IT-support vid utskrivningar i COSMIC?

- ☒ Aldrig
- ☐ Ibland
- ☐ Ofta
- ☐ Nästan alltid

Har du upplevt att patienter fått fel eller ofullständig information vid utskrivning?

- ☐ Ja
- ☐ Nej
- ☒ Vet inte

Om ja på tidigare fråga, vad berodde det på?

Har du upplevt dubbelarbete (t.ex. behövt skriva in samma sak i flera system)?

- ☐ Ja
- ☐ Nej
- ☒ Vet inte

Vilka moment upplever du som mest tidskrävande vid utskrivningsprocessen i COSMIC?

Upplever du något problem med COSMIC vid utskrivningsprocessen, i så fall vad?

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COSMIC – användarundersökning

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Vilken yrkesroll har du?

- ☐ Läkare
- ☒ Sjuksköterska
- ☐ IT - support
- ☐ Farmaceut
- ☐ Other: _____

Hur länge har du arbetat med COSMIC?

- ☐ < 1 år
- ☐ 1 - 3 år
- ☐ 4 - 6 år
- ☒ > 6 år

Hur ofta använder du COSMIC?

- ☒ Vardagligen
- ☐ Några gånger i veckan
- ☐ Sällan

Hur upplever du att utskrivningsprocessen fungerar i COSMIC idag?

- ☐ Mycket bra
- ☐ Ganska bra
- ☐ Varken eller
- ☒ Ganska dåligt
- ☐ Mycket dåligt

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☒ Skriver epikris

☒ Uppdaterar vårdplan

☐ Skickar vårdbegäran / SVPL

☐ Hanterar läkemedelslista

☐ Other:

Hur lång tid tar utskrivningen vanligtvis att genomföra i COSMIC?

Lång

Är det lätt att hitta och använda rätt funktioner i COSMIC vid utskrivning (t.ex. epikris, vårdplan, läkemedel)?

☐ Ja, alltid.

☒ Oftast

☐ Sällan

☐ Nej det är svårt

Hur ofta behöver du be om hjälp från kollegor eller IT-support vid utskrivningar i COSMIC?

- ☐ Aldrig
- ☒ Ibland
- ☐ Ofta
- ☐ Nästan alltid

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- ☐ Ja
- ☒ Nej
- ☐ Vet inte

Om ja på tidigare fråga, vad berodde det på?

Har du upplevt dubbelarbete (t.ex. behövt skriva in samma sak i flera system)?

- ☒ Ja
- ☐ Nej
- ☐ Vet inte

Vilka moment upplever du som mest tidskrävande vid utskrivningsprocessen i COSMIC?

Att vi är så många yrkesroller som ska tycka till, bhl, ssk, läkare, AT och FT. Tar lång tid att få till alla bedömningar.

Upplever du något problem med COSMIC vid utskrivningsprocessen, i så fall vad?

Vi har idag en standard vårdplan vid samordningsärenden, som borde uppdateras, den är för lång och med för många sökord. Den vårdplanen skulle kunna hjälpa oss mer med rätt sökord, så att det blir som en checklista att hålla sig till under hela utskrivningprocessen. Man behöver hjälp med att komma ihåg vad som ska göras under processen, men utan att behöva gå in i ett långt dokument. Man behöver mer utbildning av de olika yrkesrollerna vilka uppgifter i processen som de ska bidra med.

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COSMIC – user survey

The purpose of this survey is to collect experiences and opinions about the discharge process in the COSMIC medical record system. The answers are anonymous and will only be used for study purposes. The survey only takes a few minutes to complete. Thank you for your participation!

What professional role do you have?

- ☐ Doctor
- ☐ Nurse
- ☐ IT support
- ☐ Pharmacist
- ☐ Other:

How long have you been working with COSMIC?

- ☐ < 1 year
- ☒ 1 - 3 years
- ☐ 4 - 6 years
- ☐ > 6 years

How often do you use COSMIC?

- ☒ Everyday
- ☐ A few times a week
- ☐ Rarely

How do you feel the discharge process works in COSMIC today?

- ☐ Very good
- ☒ Goodish
- ☐ Neither or
- ☐ Pretty bad
- ☐ Very bad

To what extent do you use COSMIC during the discharge process of a patient?

- ☐ The entire process takes place in COSMIC
- ☒ Some parts are made in other systems
- ☐ Mostly on paper or by phone

What steps do you take in COSMIC when discharging a patient?

- ☐ Writing an epicrisis
- ☐ Updating care plan
- ☒ Sending care request / SVPL
- ☐ Manages medication list
- ☐ Other:

How long does the discharge usually take to complete in COSMIC?

-

Is it easy to find and use the right functions in COSMIC when discharging (e.g. epicrisis, care plan, medications)?

- ☒ Yes, always.
- ☐ Most often
- ☐ Rarely
- ☐ No, it's difficult.

How often do you need to ask for help from colleagues or IT support when writing in COSMIC?

- ☐ Never
- ☒ Sometimes
- ☐ Often
- ☐ Almost always

Have you experienced that patients received incorrect or incomplete information upon discharge?

- ☒ Yes
- ☐ No
- ☐ Don't know

If yes to the previous question, what was the reason?

Often misunderstanding or miscommunication

Have you experienced duplication of work (e.g. having to enter the same thing into multiple systems)?

- ☒ Yes
- ☐ No
- ☐ Don't know

What elements do you find most time-consuming in the COSMIC discharge process?

.....

Are you experiencing any problems with COSMIC during the discharge process, if so, what?

.....

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What professional role do you have?

- ☐ Doctor
- ☒ Nurse
- ☐ IT support
- ☐ Pharmacist
- ☐ Other:

How long have you been working with COSMIC?

- ☐ < 1 year
- ☐ 1 - 3 years
- ☐ 4 - 6 years
- ☒ > 6 years

How often do you use COSMIC?

- ☒ Everyday
- ☐ A few times a week
- ☐ Rarely

How do you feel the discharge process works in COSMIC today?

- ☐ Very good
- ☒ Goodish
- ☐ Neither or
- ☐ Pretty bad
- ☐ Very bad

To what extent do you use COSMIC during the discharge process of a patient?

- ☐ The entire process takes place in COSMIC
- ☒ Some parts are made in other systems
- ☐ Mostly on paper or by phone

What steps do you take in COSMIC when discharging a patient?

- ☒ Writing an epicrisis
- ☒ Updating care plan
- ☐ Sending care request / SVPL
- ☒ Manages medication list
- ☒ Other: Communicates with the municipality/private actors/health center

How long does the discharge usually take to complete in COSMIC?

Difficult to specify. The discharge process lasts from the time the patient is registered in Cosmic Link until the patient goes home. We are in continuous contact with the municipality and other stakeholders, updating the status so that other stakeholders can follow the patient, etc. On the day of discharge, the time depends on how much has been prepared beforehand. The actual discharge from the occupancy list is not time-consuming.

Is it easy to find and use the right functions in COSMIC when discharging (e.g. epicrisis, care plan, medications)?

- ☐ Yes, always.
- ☒ Most often
- ☐ Rarely
- ☐ No, it's difficult.

How often do you need to ask for help from colleagues or IT support when writing in COSMIC?

- ☒ Never
- ☐ Sometimes
- ☐ Often
- ☐ Almost always

Have you experienced that patients received incorrect or incomplete information upon discharge?

- ☒ Yes
- ☐ No
- ☐ Don't know

If yes to the previous question, what was the reason?

Often lacking information from doctors.

Have you experienced duplication of work (e.g. having to enter the same thing into multiple systems)?

- ☒ Yes
- ☐ No
- ☐ Don't know

What elements do you find most time-consuming in the COSMIC discharge process?

It is difficult to answer your questions as it is a bit unclear whether you are referring to cosmic itself or cosmic Link which is mainly used for the discharge process. It is in Cosmic Link that most of the discharge process takes place. If you continuously move the discharge process forward during the period of care and everyone does their part, the discharge process is not so time-consuming. It is more about guidelines, the design of notes/headings and confidentiality which makes the discharge process cumbersome. Many doctors also have difficulty with the discharge note, which means that it can lag behind. It is more about the content of the note and ignorance.

Are you experiencing any problems with COSMIC during the discharge process, if so, what?

It would have been easier if, for example, you could update the Status for general care directly in Cosmic Link instead of going into Status in the medical record and retrieving all search terms. It also feels like duplicate work to document/inform in both general messages in Cosmic Link and in the general care plan for discharge. For example, booked transport, who meets the patient, what time, what aids are sent with them, etc. Some actors still use Cosmic messenger as part of the discharge process, which only becomes messy and duplicate work as it is easier if everyone communicates only in Cosmic Link.

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What professional role do you have?

- ☐ Doctor
- ☐ Nurse
- ☐ IT support
- ☐ Pharmacist
- ☒ Other: Assistant nurse

How long have you been working with COSMIC?

- ☐ < 1 year
- ☐ 1 - 3 years
- ☐ 4 - 6 years
- ☒ > 6 years

How often do you use COSMIC?

- ☒ Everyday
- ☐ A few times a week
- ☐ Rarely

How do you feel the discharge process works in COSMIC today?

- ☐ Very good
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- ☐ Writing an epicrisis
- ☒ Updating care plan
- ☒ Sending care request / SVPL
- ☐ Manages medication list
- ☒ Other: most of it,

How long does the discharge usually take to complete in COSMIC?

.....

Is it easy to find and use the right functions in COSMIC when discharging (e.g. epicrisis, care plan, medications)?

- ☐ Yes, always.
- ☒ Most often
- ☐ Rarely
- ☐ No, it's difficult.

How often do you need to ask for help from colleagues or IT support when writing in COSMIC?

- ☒ Never
- ☐ Sometimes
- ☐ Often
- ☐ Almost always

Have you experienced that patients received incorrect or incomplete information upon discharge?

- ☐ Yes
- ☐ No
- ☒ Don't know

If yes to the previous question, what was the reason?

.....

Have you experienced duplication of work (e.g. having to enter the same thing into multiple systems)?

- ☐ Yes
- ☐ No
- ☒ Don't know

What elements do you find most time-consuming in the COSMIC discharge process?

.....

Are you experiencing any problems with COSMIC during the discharge process, if so, what?

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- ☐ Goodish
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- ☒ Writing an epicrisis
- ☒ Updating care plan
- ☐ Sending care request / SVPL
- ☐ Manages medication list
- ☐ Other:

How long does the discharge usually take to complete in COSMIC?

Long

Is it easy to find and use the right functions in COSMIC when discharging (e.g. epicrisis, care plan, medications)?

- ☐ Yes, always.
- ☒ Most often
- ☐ Rarely
- ☐ No, it's difficult.

How often do you need to ask for help from colleagues or IT support when writing in COSMIC?

- ☐ Never
- ☒ Sometimes
- ☐ Often
- ☐ Almost always

Have you experienced that patients received incorrect or incomplete information upon discharge?

- ☐ Yes
- ☒ No
- ☐ Don't know

If yes to the previous question, what was the reason?

.....

Have you experienced duplication of work (e.g. having to enter the same thing into multiple systems)?

- ☒ Yes
- ☐ No
- ☐ Don't know

What elements do you find most time-consuming in the COSMIC discharge process?

That we are so many professional roles that have to be considered, BHL, SSK, doctors, AT and FT. It takes a long time to get all the assessments.

Are you experiencing any problems with COSMIC during the discharge process, if so, what?

We currently have a standard care plan for coordination cases, which should be updated, it is too long and with too many keywords. That care plan could help us more with the right keywords, so that it becomes like a checklist to stick to during the entire discharge process. You need help remembering what to do during the process, but without having to go into a long document. You need more training of the different professional roles and what tasks in the process they should contribute.

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